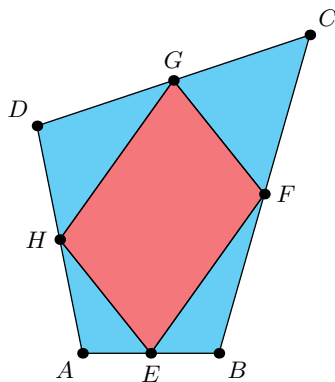


0.0.1

E , F , G og H er de respektive midtpunktene til sidene til $\square ABCD$.

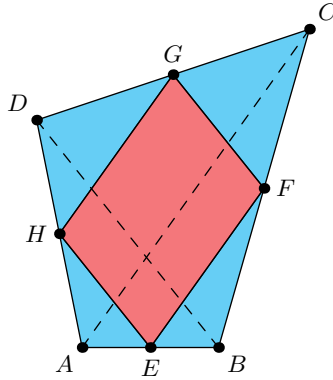
a) Vis at $\square EFGH$ er et parallelogram.

b) Vis at $A_{\square EFGH} = \frac{1}{2}A_{\square ABCD}$



Merk: Sammenhengen fra oppgave a) kalles **Varignons teorem**.

Svar



- a) $\triangle ACD \sim \triangle HGD$ fordi $\angle ADC = \angle HDG$ og $\frac{AD}{HD} = \frac{DC}{DG} = 2$. Tilsvarende er $\triangle ABC \sim \triangle EBF$. Dermed er både HG og EF parallelle med AC . Tilsvarende kan vi vise at HE og GF er parallelle med BD . Dermed er $\square EFGH$ et parallelogram.
- b) Vi har at

$$2A_{\square ABCD} = A_{\triangle ACD} + A_{\triangle ABC} + A_{\triangle ABD} + A_{\triangle BCD} \quad (1)$$

og at

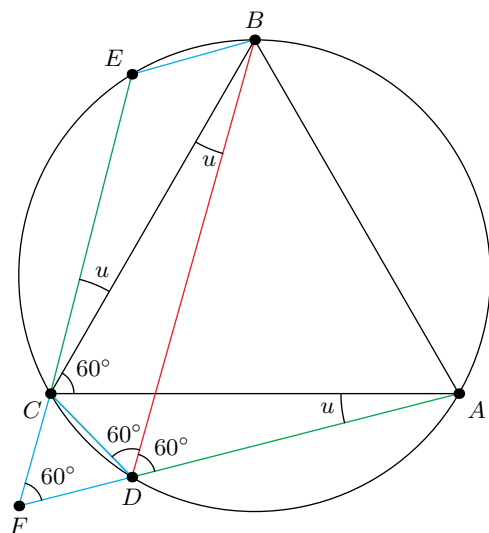
$$A_{\square EFGH} = A_{\square ABCD} - A_{\triangle HGD} - A_{\triangle EBF} - A_{\triangle AEH} - A_{\triangle FCH} \quad (2)$$

Av formlikheten vi viste i oppgave a), vet vi at

$$A_{\triangle HGD} = \left(\frac{1}{2}\right)^2 A_{\triangle ACD} = \frac{1}{4} A_{\triangle ACD}$$

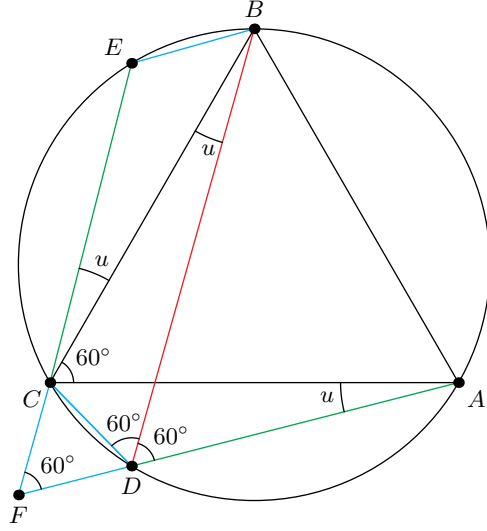
Tilsvarende sammenheng har vi mellom de andre trekantene fra (1) og trekantene fra (2), og dermed er

$$A_{\square EFGH} = A_{\square ABCD} - \frac{1}{4} \cdot 2A_{\square ABCD} = \frac{1}{2} A_{\square ABCD}$$



Da de er periferivinkler som spenner over buer med lik lengde, er $\angle ADB = \angle BDC = \angle ACB = 60^\circ$. Tilsvarende er $\angle CAD = \angle CBD = u$. Vi plasserer F på forlengelsen av AD slik at $\angle DFC = 60^\circ$. Da er $\triangle CFD$ likesidet. Videre plasserer vi E slik at $\triangle BCE \cong \triangle CAD$. Av vinkelsummen til $\triangle CAD$, er $\angle DCA = 60^\circ - u$, og følgelig er $\angle FCE = 180^\circ$. Videre har vi at $\angle EBD = \angle EBC + \angle CBD = 60^\circ - u + u = 60^\circ$. Følgelig er $\square FDEB$ et parallelogram ($\angle CEB = \angle BDF = 120^\circ$), og da er

$$BD = EF = FC + CE = CD + DA$$



Since they are peripheral angles spanning arcs of equal length, $\angle ADB = \angle BDC = \angle ACB = 60^\circ$. Similarly, $\angle CAD = \angle CBD = u$. We place F on the extension of AD such that $\angle DFC = 60^\circ$. Then, $\triangle CFD$ is equilateral. Furthermore, we place E such that $\triangle BCE \cong \triangle CAD$. From the angle sum of $\triangle CAD$, $\angle DCA = 60^\circ - u$, and consequently, $\angle FCE = 180^\circ$. Additionally, we have $\angle EBD = \angle EBC + \angle CBD = 60^\circ - u + u = 60^\circ$. Consequently, quadrilateral $FDEB$ is a parallelogram ($\angle CEB = \angle BDF = 120^\circ$), and thus

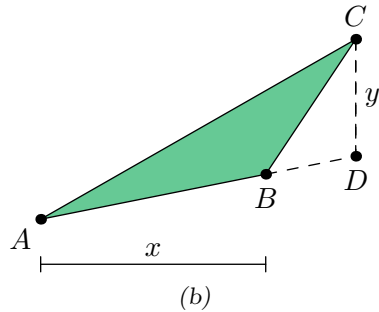
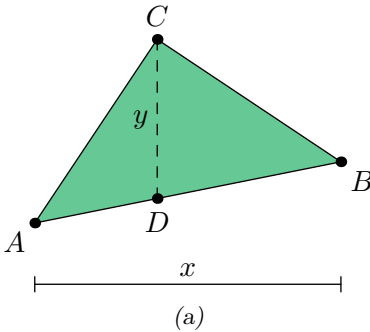
$$BD = EF = FC + CE = CD + DA.$$

Gruble 1

Gitt en trekant $\triangle ABC$, hvor horisontalavstanden mellom A og B er x . Videre er $y = CD$ vertikalavstanden mellom C og AB , eller mellom C og forlengelsen av AB .

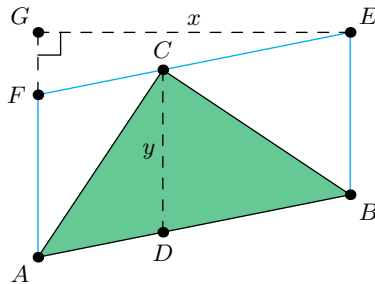
Vis at

$$A_{\triangle ABC} = \frac{xy}{2}$$



Svar

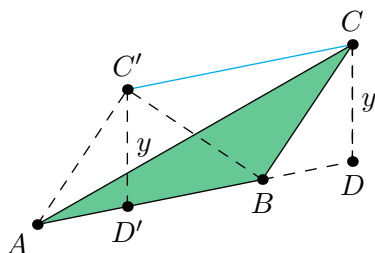
Tilfellet hvor D ligger på AB



Vi danner parallelogrammet $\square ABEF$, hvor $AF \parallel EB \parallel CD$. Med AB som grunnlinje, har $\square ABEF$ og $\triangle ABC$ lik høyde, og dermed er

$$A_{\triangle ABC} = \frac{1}{2} A_{\square ABEF} = \frac{1}{2} AF \cdot GE = \frac{yx}{2}$$

Tilfellet hvor D ligger på forlengelsen av AB



C' er C parallellforskjøvet langs en linje som er parallell med AB , slik at D' ligger på AB . Da vet vi at

$$A_{\triangle ABC'} = \frac{xy}{2}$$

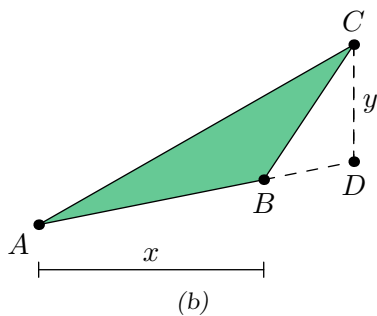
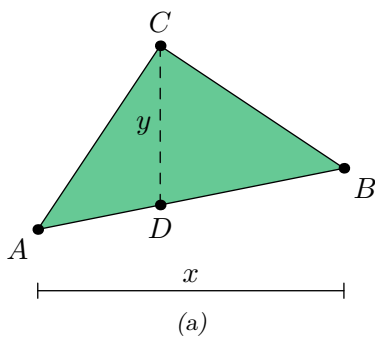
Med AB som grunnlinje, har $\triangle ABC'$ og $\triangle ABC$ lik høyde, og dermed likt areal. Følgelig er

$$A_{\triangle ABC} = A_{\triangle ABC'} = \frac{xy}{2}$$

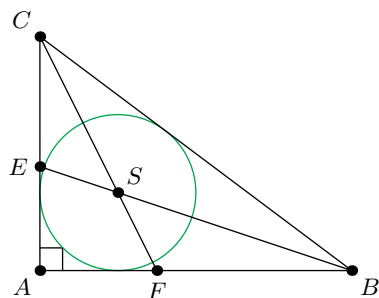
Given a triangle $\triangle ABC$, where the horizontal distance between A and B is x . Furthermore, let $y = CD$ denote the vertical distance from C to the line segment AB , or to the extension of AB .

Show that

$$A_{\triangle ABC} = \frac{xy}{2}.$$

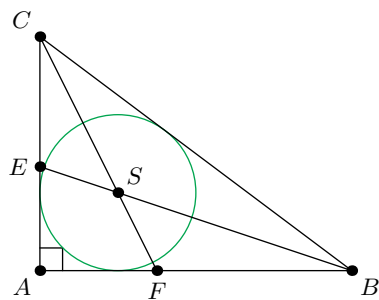


Sirkelen har radius r . Vis at $4BC \cdot r = \sqrt{2}BE \cdot CF$



Let r denote the radius of the circle. Show that

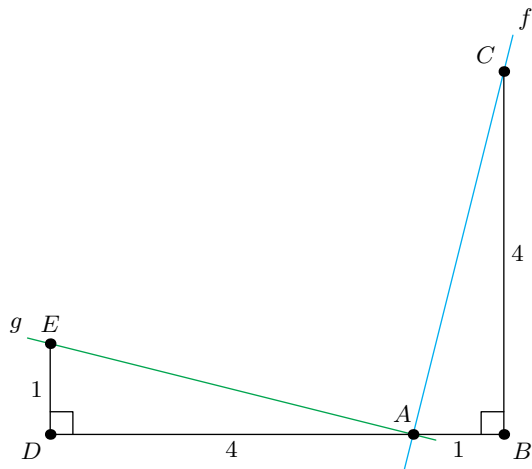
$$4BC \cdot r = \sqrt{2}BE \cdot CF$$



- a) Given the functions $f(x) = 4x$ and $g(x) = -\frac{1}{4}x$. Show that the graph of f and the graph of g are perpendicular.
- b) Given the functions $f(x) = ax + b$ and $g(x) = -\frac{1}{a}x + c$. Show that the graph of f and the graph of g are perpendicular.

a) **Method 1**

The graphs of f and g intersect when $x = 0$. We set $A = (0, 0)$, $B = (1, 0)$, and $C = (1, f(1)) = (1, 4)$. Furthermore, we set $D = (-4, 0)$ and $E = (-4, g(-4)) = (-4, 1)$. Since $DE = AB$ and $DA = BC$, and $\angle ADE = \angle CBA$, the triangles $\triangle EDA \sim \triangle ABC$.



Method 2

The graphs of f and g clearly intersect when $x = 0$. We set $A = (0, 0)$, $B = (1, 0)$, and $C = (1, f(1)) = (1, 4)$. Let E be the point on the graph of g with the same y -value as C . At E ,

$$\begin{aligned} g(x) &= 4 \\ -\frac{1}{4}x &= 4 \\ x &= -16 \end{aligned}$$

Thus, $E = (-16, 4)$. We set $D = (-16, 0)$. By the Pythagorean theorem on $\triangle AED$, we have

$$EA^2 = ED^2 + AD^2 = 4^2 + 16^2 = 16 + 16^2 = 16(1 + 16) = 16 \cdot 17$$

By the Pythagorean theorem on $\triangle BAC$, we have

$$AC^2 = AB^2 + BC^2 = 1^2 + 4^2 = 17$$

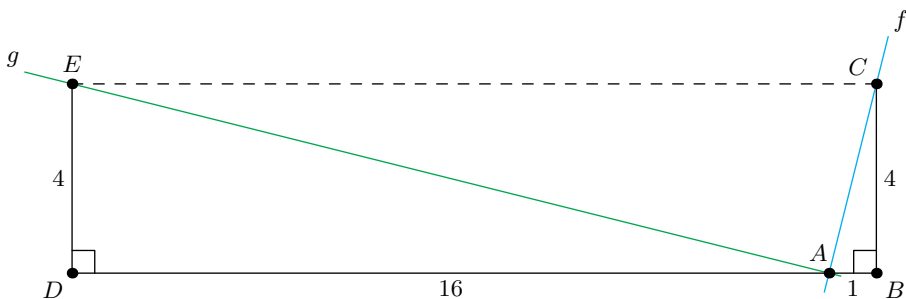
If the graphs of f and g are perpendicular to each other, then $\angle CAE = 90^\circ$, and $\triangle AEC$ must satisfy the Pythagorean theorem. We know that $EC = BD = 17$, and thus we must require that

$$EC^2 = AC^2 + EA^2 = 17^2$$

We have

$$AC^2 + EA^2 = 17 + 16 \cdot 17 = 17(1 + 16) = 17^2$$

Thus, $\triangle AEC$ is a right triangle.



b) We assume $a > 1$. If $f = g$, we have

$$ax + b = -\frac{1}{a}x + c$$

$$x = \frac{a(b - c)}{a^2 + 1}$$

This equation is guaranteed to have a solution, and thus we know that f and g intersect at a point A . We place B and D on the same horizontal line as A such that $AD = BD = 1$. Furthermore, we place E and C on the graphs of g and f , respectively, such that $DA \perp ED$ and $AB \perp BC$. Then, $DE = \frac{1}{a}$ and $BC = a$. We know that in $\triangle ADE$, the longest leg is AD and the shortest leg is DE , and in $\triangle ABC$, the longest leg is BC and the shortest leg is AB . Furthermore,

$$\frac{DE}{AD} = \frac{AB}{BC} = \frac{1}{a}$$

Since $\angle ADE = \angle CBA$, it follows that $\triangle ADE \sim \triangle CBA$, and thus

$$\begin{aligned} \angle CAE &= 180^\circ - \angle EAD - \angle BAC \\ &= 180^\circ - \angle EAD - (90^\circ - \angle EAD) \\ &= 90^\circ \end{aligned}$$

The explanations for the cases where $0 < a < 1$ or $a < 0$ are almost identical to the one we have given for $a > 1$.

